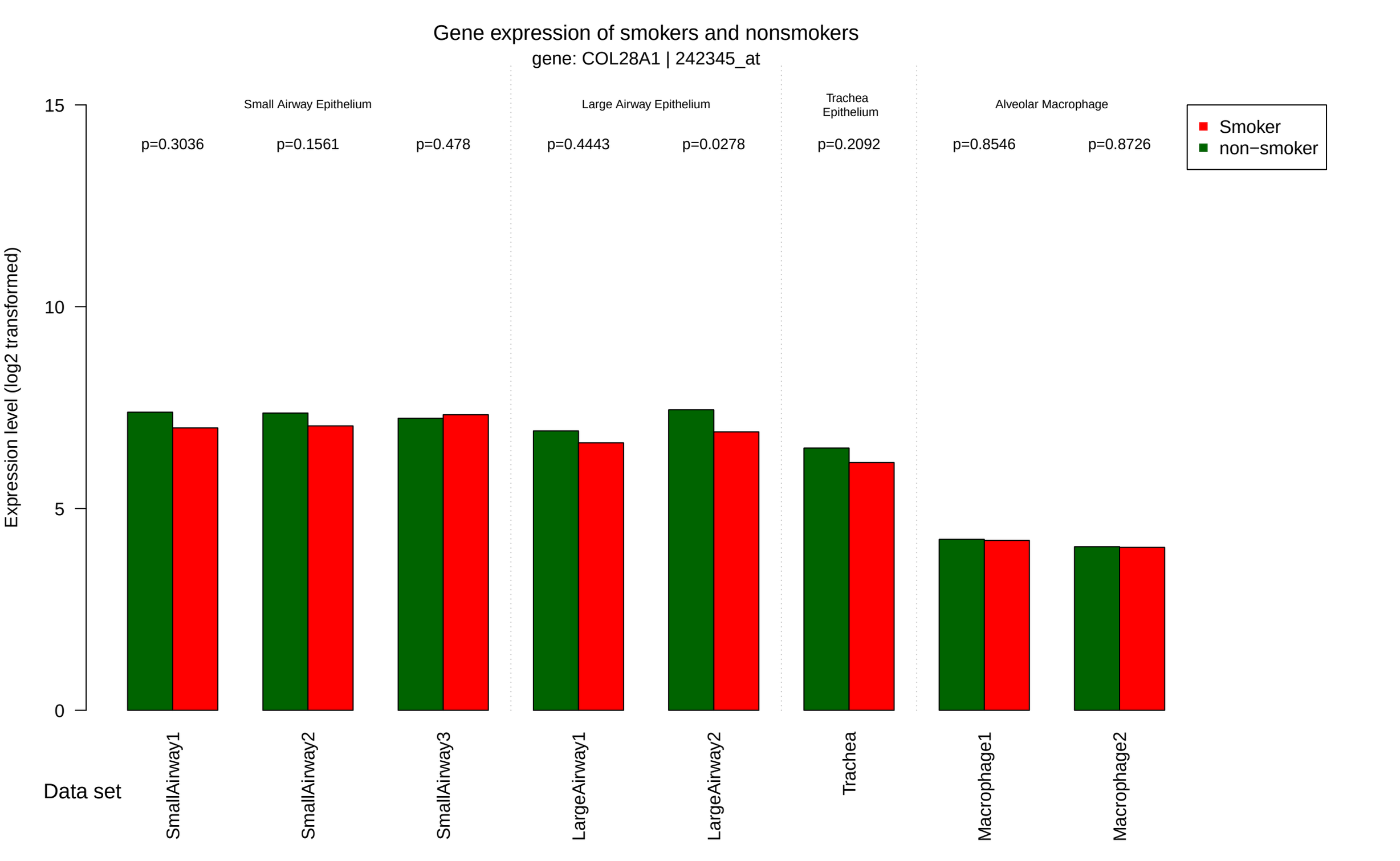
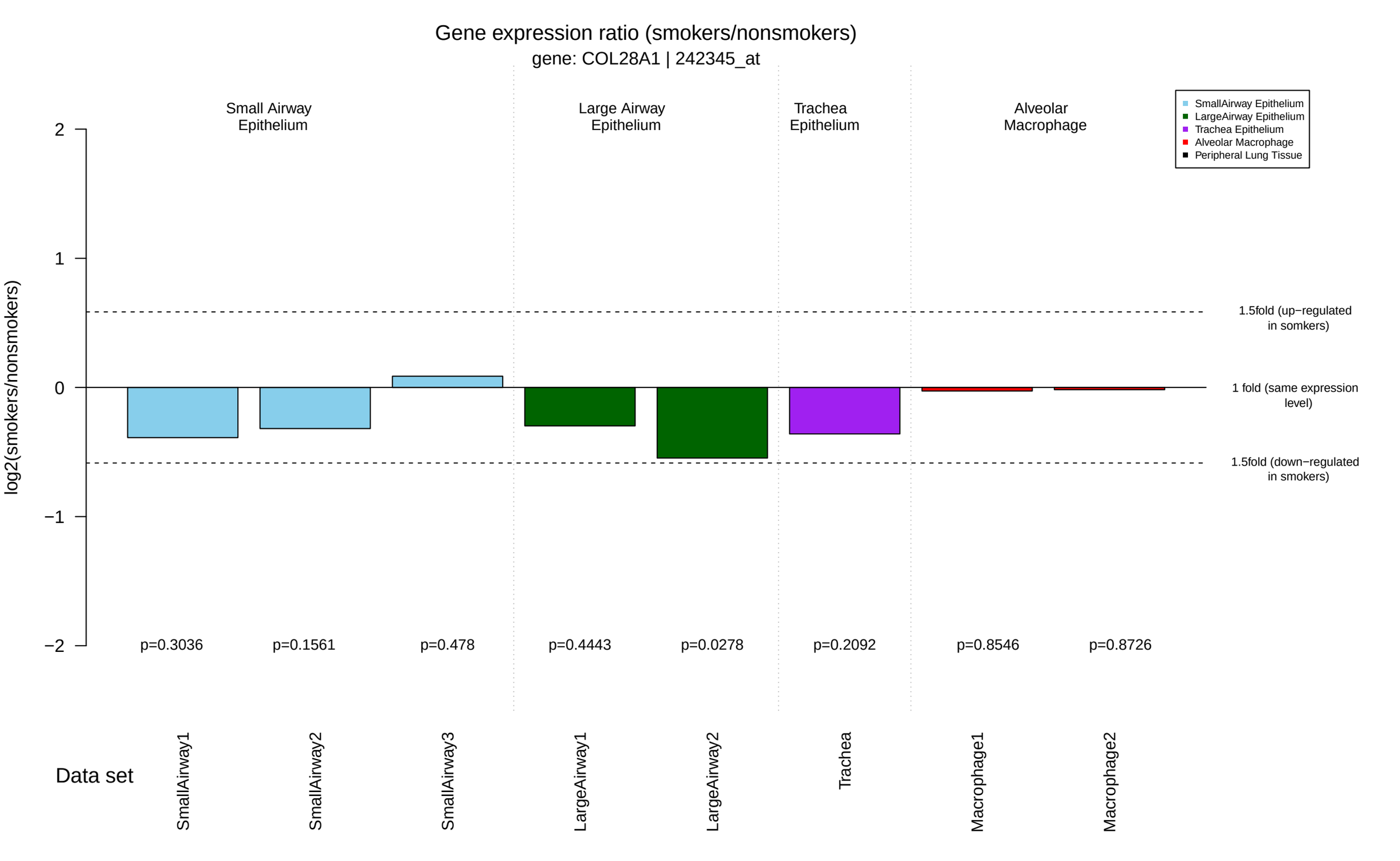
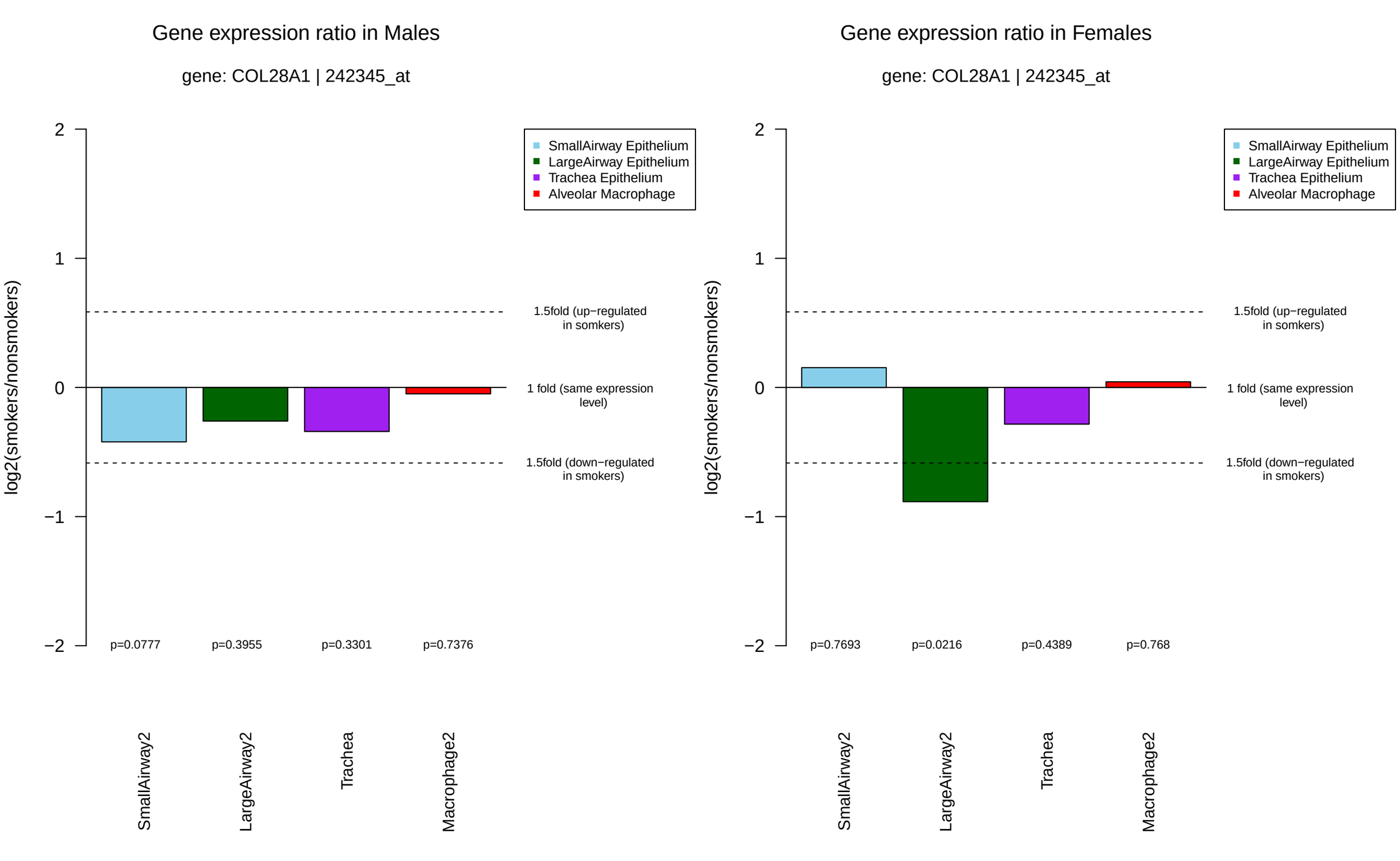
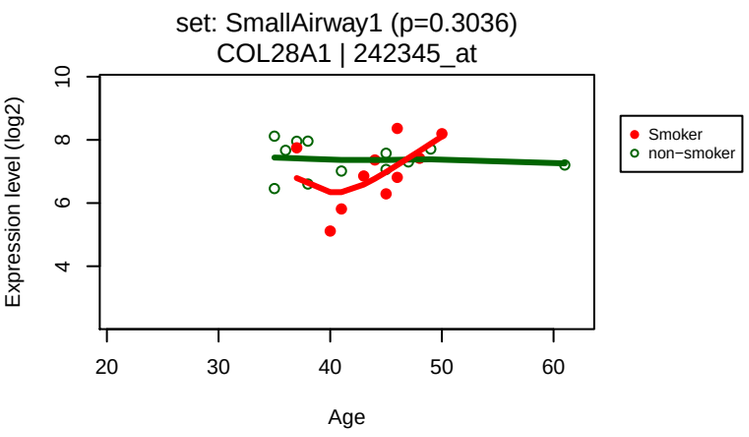
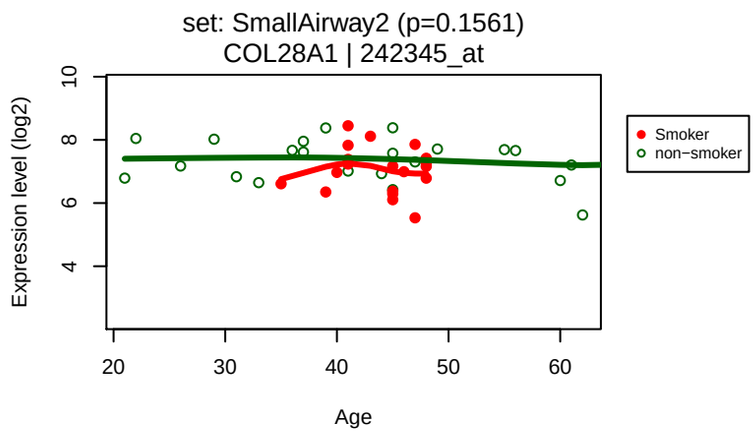
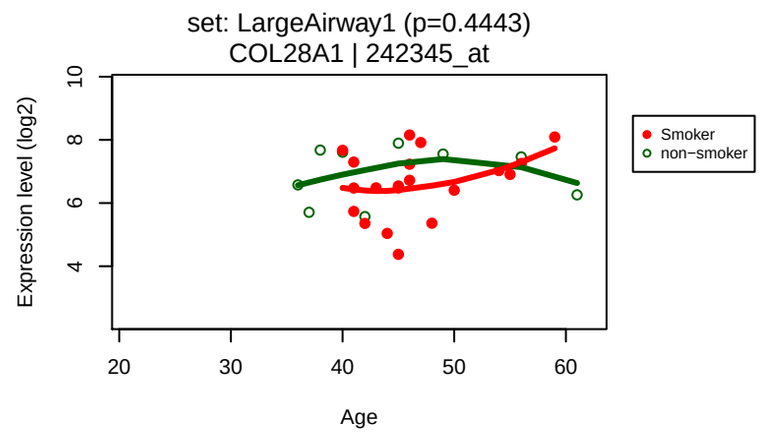

Gene expression of smokers and nonsmokers
gene: COL28A1 | 242345_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3036
p=0.1561
p=0.478
p=0.4443
p=0.0278
p=0.2092
p=0.8546
p=0.8726
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

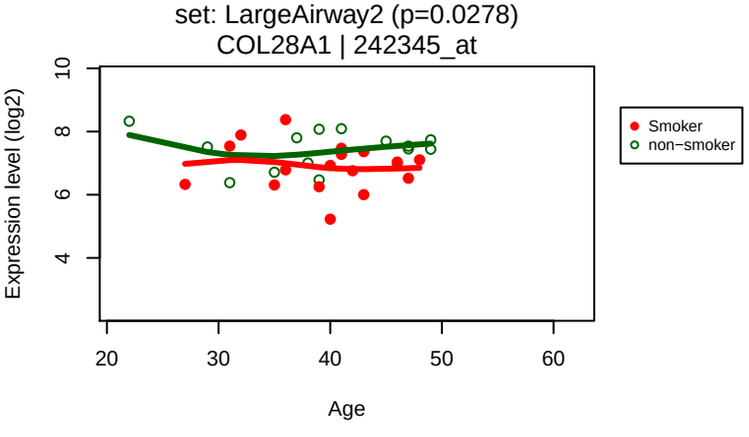

Gene expression ratio (smokers/nonsmokers)
gene: COL28A1 | 242345_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3036
p=0.1561
p=0.478
p=0.4443
p=0.0278
p=0.2092
p=0.8546
p=0.8726
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

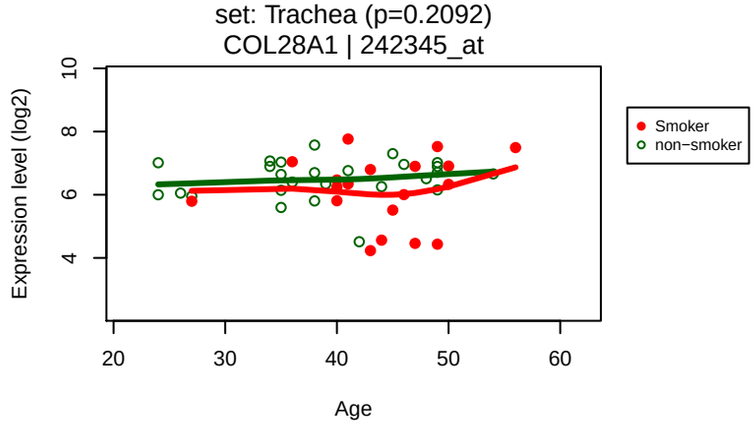

Gene expression ratio in Males
gene: COL28A1 | 242345_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0777
p=0.3955
p=0.3301
p=0.7376
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: COL28A1 | 242345_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7693
p=0.0216
p=0.4389
p=0.768
SmallAirway2
LargeAirway2
Trachea
Macrophage2

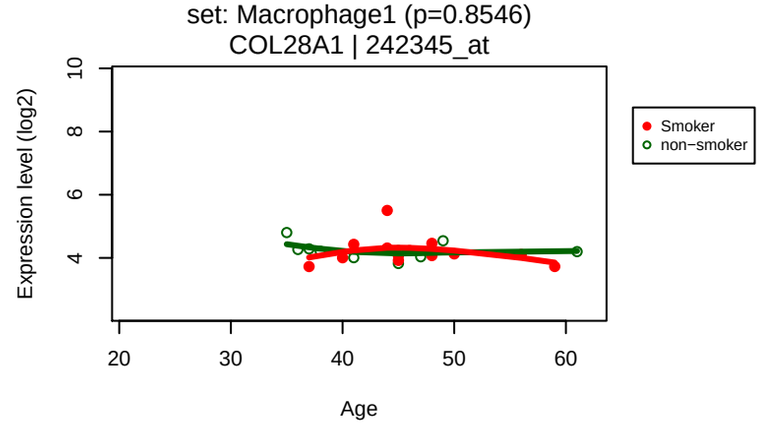

set: SmallAirway1 (p=0.3036)
COL28A1 | 242345_at
Expression level (log2)
Age
Smoker
non−smoker

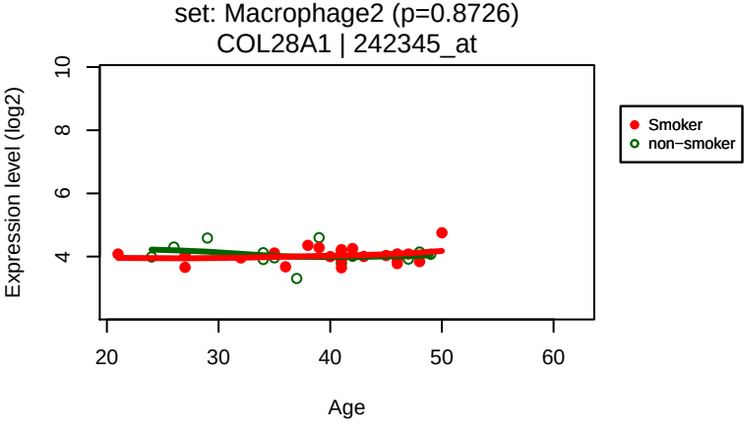

set: SmallAirway2 (p=0.1561)
COL28A1 | 242345_at
Expression level (log2)
Age
Smoker
non−smoker

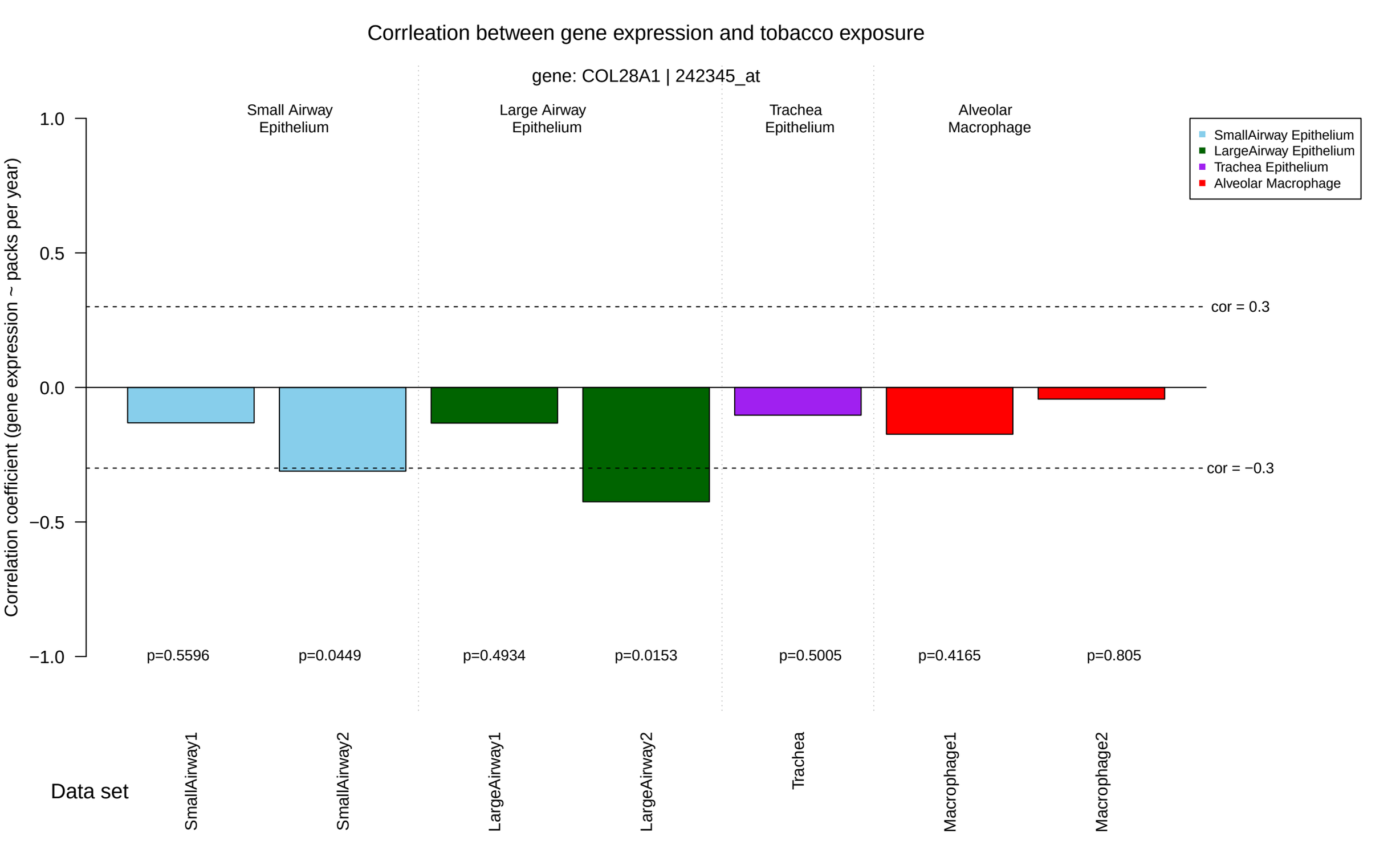

set: LargeAirway1 (p=0.4443)
COL28A1 | 242345_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0278)
COL28A1 | 242345_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2092)
COL28A1 | 242345_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8546)
COL28A1 | 242345_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8726)
COL28A1 | 242345_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: COL28A1 | 242345_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5596
p=0.0449
p=0.4934
p=0.0153
p=0.5005
p=0.4165
p=0.805
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2